

Queue-Aware versus Resource-Based Autoscaling on Amazon EKS: An Experimental Comparison of KEDA and HPA under Poisson and Bursty Workloads

Lucas Melo Rocha

NUSP 13680400

PSI5120 – Topics in Cloud Computing

Polytechnic School, University of São Paulo

Abstract—Message-driven applications can scale from resource pressure observed inside containers or from demand waiting outside them. This paper compares these two approaches on Amazon Elastic Kubernetes Service. The resource-based treatment uses the Kubernetes Horizontal Pod Autoscaler with CPU utilization, while the queue-aware treatment uses Kubernetes Event-Driven Autoscaling with the backlog of an Amazon Simple Queue Service queue. Both treatments execute the same worker image and are constrained to one through four replicas on the same cluster. Reproducible traces represent homogeneous Poisson arrivals and a two-state Markov-modulated Poisson process with the same target mean but different burstiness. The paired experiment evaluates end-to-end latency p95 and ready pod-seconds as primary outcomes, with backlog, queue wait, scale-out delay, and drain time as secondary outcomes. Inference uses one aggregate per run and paired bootstrap intervals instead of treating messages as independent replicates. Across nine Poisson and eight MMPP pairs, KEDA had 47.8% and 29.9% less ready-capacity exposure, respectively. This came with geometric-mean p95 latency ratios of 2.03 and 2.66 relative to HPA. MMPP latency was sensitive to a realized confounding between order and campaign time. Thus, the tested queue-aware configuration favored lower ready capacity rather than lower tail latency; pod-seconds do not represent direct cloud cost. The complete lifecycle is automated with least-privilege identities, raw-data retention, and cleanup auditing.

Index Terms—Amazon EKS, autoscaling, HPA, KEDA, Kubernetes, MMPP, SQS.

I. INTRODUCTION

Elasticity is a central promise of cloud computing, but a controller can react only to demand that its metric exposes. A CPU-based controller observes work after a Pod receives it. A queue-based controller can observe waiting work before a consumer starts processing it. This distinction may be immaterial under smooth traffic, yet important when short overload periods build a backlog faster than resource metrics and control loops react.

Asynchronous services make this choice especially visible. A request-response autoscaler often observes traffic at an ingress or utilization at a server. A queue worker instead separates admission from execution: the broker accepts a message, the message waits for a consumer, CPU work

executes, and a result is published later. End-to-end time can therefore be decomposed as

$$T_{e2e} = T_{\text{queue}} + T_{\text{service,wall}}. \quad (1)$$

Scaling can reduce the first term by adding consumers and can affect the second through CPU contention and throttling. Its benefit arrives only after sensing, reconciliation, scheduling, image startup, and readiness delays. Short bursts may finish before this complete control path reacts.

The operational decision is consequently not just which software component to install. It combines signal placement with target calibration. A queue target states how much waiting work is tolerated per replica; a CPU target states how much requested compute should be occupied before another replica is desired. Equal replica bounds do not make those operating points equivalent. This study compares two frozen, deployable configurations and then examines the mechanisms behind their observed capacity-latency trade-off.

Kubernetes provides the Horizontal Pod Autoscaler (HPA), which adjusts a workload’s replica count from resource, custom, or external metrics [1]. Kubernetes Event-Driven Autoscaling (KEDA) integrates external event sources with the Kubernetes scaling API. Its Amazon SQS scaler uses visible and, by default, in-flight messages to derive desired capacity [2]. Both ultimately manipulate the same Deployment, but their signals represent different points in the request path.

Prior work has examined HPA behavior, alternative resource metrics, traffic prediction, and adaptive policies [3], [4], [5], [6], [7]. Event-queue autoscaling has also been studied as a tail-latency and resource-efficiency problem [8]. We are not aware of a published real-cloud comparison that combines frozen paired traces, HPA/CPU, KEDA/SQS, memoryless and correlated arrivals, run-level inference, and retained raw controller observations. This is a scoped positioning claim, not a systematic review.

This paper asks: how do CPU-based HPA and SQS-backlog-based KEDA differ in latency, backlog control, scaling responsiveness, and ready-capacity exposure when offered

loads have similar means but different burstiness? The study makes five contributions:

- a paired comparison that changes the autoscaling policy while holding the application, cluster, queues, traces, and replica limits constant;
- a reproducible Poisson and two-state MMPP workload generator that stores every arrival offset and CPU service demand before execution;
- an analysis based on run-level aggregates, paired ratios, and bootstrap confidence intervals;
- mechanism evidence that separates observed desired-replica timing from subsequent Pod readiness and quantifies replica-level occupancy;
- a guarded and auditable EKS lifecycle that includes least-privilege IAM roles, cost controls, and resource deletion checks.

II. BACKGROUND AND RELATED WORK

A. Autoscaling as Delayed Feedback

A reactive autoscaler is a sampled-data feedback system. The application first changes the monitored state; a metrics path samples and exports that state; a controller computes a desired replica count; the Deployment creates Pods; and the scheduler, container runtime, and readiness gate make capacity usable. Let d_s , d_c , and d_a denote sensing, controller, and actuation delay. New capacity cannot affect a message before approximately $d_s + d_c + d_a$. Metric freshness, polling cadence, controller tolerance, stabilization windows, image cache state, and node headroom all contribute.

This decomposition matters for policy interpretation. A demand signal may be upstream and semantically direct yet sampled slowly. A downstream CPU signal may observe demand later but cross an aggressive target quickly. Conversely, keeping more idle capacity can dominate signal quality for short traces. The experiment records desired and ready replicas every five seconds to observe the combined decision and actuation path, but it does not instrument every internal controller event.

B. Resource-Based Horizontal Scaling

For CPU utilization, HPA periodically reads metrics from the Kubernetes Metrics API. In simplified form, its replica recommendation is

$$c_{t+1} = \left\lceil c_t \frac{u_{\text{observed}}}{u_{\text{target}}} \right\rceil, \quad (2)$$

where c_t is the current replica count and utilization is measured relative to each container's CPU request [1]. Tolerance, unavailable metrics, stabilization, and rate policies modify the practical behavior. HPA therefore observes a consequence of demand rather than the queue itself.

Nguyen et al. analyze elastic container orchestration with HPA and show the importance of resource configuration and workload conditions [3]. Casalicchio and Perciballi compare relative and absolute container metrics [7]. Balla et al. and Toka et al. propose adaptive approaches beyond a fixed

reactive threshold [6], [5]. Phuc et al. add traffic awareness in an edge-computing setting [4]. These studies motivate careful metric selection and reproducible load control.

C. Queue-Aware Event-Driven Scaling

KEDA exposes an external metric and creates an HPA for the target workload. For the SQS scaler used here, the observed demand is

$$q_t = q_{\text{visible},t} + q_{\text{inflight},t}. \quad (3)$$

With a target of b messages per Pod, the central recommendation is approximately $\lceil q_t/b \rceil$, subject to polling, HPA behavior, and the configured replica bounds [2]. SQS counters are approximate, and AWS advises interpreting their semantics rather than treating them as exact transactional values [9].

Queue-aware control can react while work is waiting, but it requires access to the queue metric and a meaningful backlog target. Ezzeddine et al. frame event queue autoscaling around tail latency and resource-efficient placement [8]. The present work narrows the comparison to two widely deployable Kubernetes policies and one shared SQS consumer.

D. Synthesis of Related Evaluations

Autoscaling surveys distinguish reactive and predictive methods, horizontal and vertical actions, and resource-, application-, and infrastructure-level signals [10]. Table I positions the studies used here by signal and evaluation emphasis. Nguyen et al. examine standard HPA behavior and resource configuration [3]. Casalicchio and Perciballi show that relative and absolute container metrics can produce different scaling behavior [7]. Balla et al. and Toka et al. adapt control beyond a fixed target, while Phuc et al. incorporate traffic observations [6], [5], [4]. Ezzeddine et al. optimize event-queue placement around tail latency and resource efficiency [8].

The comparison highlights three methodological choices in this work. First, every policy pair receives exactly the same pre-generated arrival and CPU-demand sequence. Second, uncertainty is computed across complete runs rather than messages. Third, the execution uses managed EKS and SQS rather than a simulator. These choices trade breadth of configurations for stronger control of offered work and direct evidence from a production cloud substrate.

E. Bursty Arrival Processes

A homogeneous Poisson process has independent exponential inter-arrival times and a constant rate λ . It is useful as a memoryless baseline, but many real workloads exhibit correlated bursts. A two-state MMPP combines a Markov chain with state-dependent Poisson rates λ_L and λ_H [11]. If transition rates are r_{LH} and r_{HL} , the stationary high-state probability is

$$\pi_H = \frac{r_{LH}}{r_{LH} + r_{HL}}, \quad (4)$$

and the stationary mean arrival rate is

Table I
POSITIONING OF RELATED AUTOSCALING EVALUATIONS

Work	Primary signal	Control emphasis	Workload emphasis	Main outcomes
Nguyen et al. [3]	CPU/resource	Kubernetes HPA	Synthetic demand	Elasticity, resources
Casalicchio et al. [7]	Relative/absolute metrics	Container scaling	Synthetic demand	Metric impact
Balla et al. [6]	Resource metrics	Adaptive policy	Variable demand	Response, efficiency
Toka et al. [5]	Multiple metrics	AI-based adaptation	Trace-driven demand	Scaling quality
Phuc et al. [4]	Traffic and resources	Traffic-aware HPA	Edge traffic	Delay, resources
Ezzeddine et al. [8]	Event queues	Placement/autoscaling	Distributed queues	Tail latency, efficiency
This study	CPU versus SQS backlog	Frozen HPA/KEDA	Paired Poisson/MMPP	p95 latency, ready pod-s

$$\bar{\lambda} = (1 - \pi_H)\lambda_L + \pi_H\lambda_H. \quad (5)$$

The experiment sets the Poisson rate and MMPP stationary mean to the same value, while preserving intervals in which MMPP demand exceeds one-Pod capacity.

III. RESEARCH METHOD

A. Questions and Outcomes

The experiment addresses three research questions:

- RQ1: Which policy provides lower per-run end-to-end latency p95 under Poisson and MMPP arrivals?
- RQ2: Which policy has fewer ready pod-seconds over arrival and drain?
- RQ3: How do the policies differ in backlog, queue wait, scale-out delay, peak replicas, and drain time?

End-to-end latency starts at the SQS service timestamp for the accepted input message and ends when the worker completes processing. Ready pod-seconds are the time integral of ready replicas from the start of arrivals until the last worker completion. These are primary outcomes. Secondary outcomes include queue wait p50, p95, and p99; maximum and integrated backlog; time to the first additional ready replica; drain time after the last accepted input; peak replicas; producer lag; duplicates; and missing results.

The primary estimand for outcome Y is the geometric mean, across paired seeds, of the within-trace KEDA/HPA ratio. It describes a multiplicative configuration effect for each workload stratum. Ready pod-seconds measure exposure to ready consumer capacity. They are not CPU-seconds, energy, node-hours, or AWS cost: the same fixed EC2 node remained allocated under both policies. Secondary estimands are exploratory and serve mainly to explain the primary trade-off.

B. Experimental Design

The design is paired and blocked by workload. Autoscaler has two levels, HPA and KEDA, and arrival process has two strata, Poisson and MMPP. Each pre-generated trace is replayed once under each policy. Treatment order is deterministically shuffled within each workload and seed. The experimental unit is a complete run, not a message.

Five paired seeds are evaluated first. Five additional seeds are required if an operationally invalid attempt occurs or if any 95% paired bootstrap interval for the two primary KEDA/HPA

ratios extends beyond plus or minus 10% of its point estimate. The experiment stops after ten paired seeds regardless of that precision criterion. Invalid attempts remain available for diagnosis and are not included as valid observations.

The realized design is best understood as a blocked paired comparison within two workload strata. Workload is not a repeatedly randomized treatment: each seed defines one Poisson trace and one independently generated MMPP trace. A descriptive ratio-of-ratios across common seeds examines whether the KEDA/HPA contrast changes with workload, but it is not treated as a confirmatory factorial interaction.

C. Architecture and Control Paths

Figure 1 separates the data, control, and measurement planes. The external replayer sends a frozen trace to the input queue. Worker Pods receive one message at a time, perform CPU work, publish a result, and then delete the input. The collector drains the result queue while an independent monitor samples both queues and Kubernetes state.

The HPA path obtains worker CPU utilization through Metrics Server. The KEDA path authenticates to SQS, reads approximate visible and in-flight counts, exports an external metric, and manages a generated HPA. Both paths ultimately write the Deployment’s desired count. Separate IAM roles for service accounts prevent the worker from reading scale metrics and prevent KEDA from consuming messages.

D. Controlled Environment

Table II summarizes the frozen configuration. Both policies share one Deployment and differ only in the active autoscaler. KEDA and the worker use separate IAM roles for service accounts. This follows least privilege and avoids placing long-lived AWS credentials in Pods [12].

The cluster has public worker networking and no NAT Gateway. It creates no load balancer, database, dashboard stack, or node autoscaler. Four workers can consume at most 1.6 vCPU by cgroup limit, leaving nominal CPU capacity for Kubernetes, KEDA, and Metrics Server on the 2-vCPU node.

Identical replica limits do not imply equivalent target aggressiveness. Mean offered CPU demand is approximately $1.6 \times 0.2 = 0.32$ CPU. HPA targets 60% of a 0.20-CPU request, or 0.12 CPU per replica. Ignoring overhead and sampling, the mean-load target therefore corresponds to $\lceil 0.32/0.12 \rceil = 3$ replicas. The 0.40-CPU limit bounds execution but is not the denominator of HPA utilization.

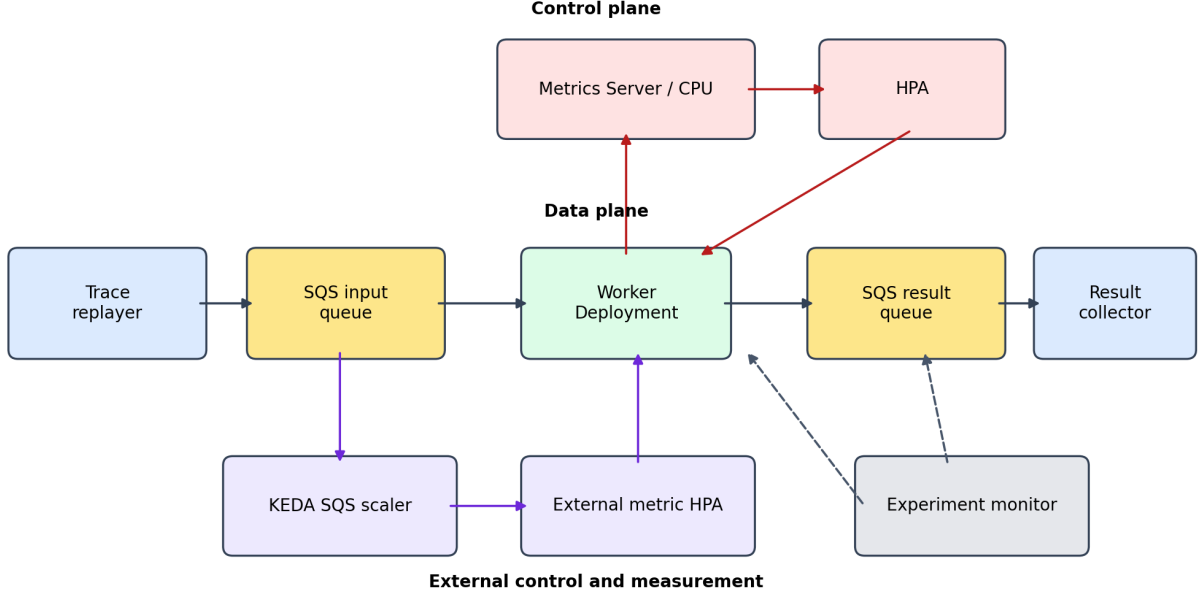


Figure 1. Experimental architecture. Gray arrows carry messages, red arrows show the CPU/HPA path, purple arrows show the SQS/KEDA path, and dashed arrows represent independent observation. Only one autoscaling path is active in a measured run.

Table II
FROZEN EXPERIMENTAL CONFIGURATION

Component	Configuration
Region	us-east-1
EKS nodes	one c7i-flex.large
Node scaling	disabled, one node fixed
Worker replicas	minimum 1, maximum 4
Worker CPU	request 200m, limit 400m
Worker memory	request 96 MiB, limit 192 MiB
HPA signal	CPU, target 60% of request
KEDA signal	visible plus in-flight SQS messages
KEDA target	5 messages per Pod
Polling interval	15 s
Scale-down stabilization	60 s
Input visibility timeout	120 s
Trace duration	180 s

KEDA instead tolerates approximately five visible-plus-in-flight messages per desired Pod. The experiment compares these frozen policy configurations; it does not claim that their targets encode the same latency or capacity objective.

E. Workload Construction

The common mean arrival rate is 1.6 messages/s. CPU service demand is exponential with mean 0.2 CPU-s and is clipped to the interval from 0.005 to 1 CPU-s. At a 0.4-CPU Pod limit, one continuously busy replica has an approximate capacity of 2 messages/s before overhead. Thus, the mean load is below one-replica capacity, while bursts can exceed it.

MMPP rates are 0.4 and 4 messages/s. Transition rates are 0.05/s from low to high and 0.10/s from high to low. Therefore $\pi_H = 1/3$ and the stationary mean is 1.6 messages/s. The chain starts in its stationary distribution. Because a 180-

second realization can differ greatly from its stationary mean, deterministic rejection sampling accepts an MMPP trace only when its count is within 10% of the expected count. The base seed, accepted arrival seed, attempt, empirical rate, offsets, and demands are retained. This rule uses only offered-load counts and no autoscaling outcome.

Short-window counts verify that equal means did not erase burstiness. Across nine traces, median maximum arrivals in 1-, 5-, 15-, and 30-second windows were 7, 17, 39, and 63 for Poisson, versus 11, 30, 68, and 103 for MMPP. The median inter-arrival coefficient of variation was 0.991 and 1.938. MMPP generated a median 85.1% of arrivals in the high state. This is close to the theoretical arrival-weighted fraction

$$P(H \mid \text{arrival}) = \frac{\pi_H \lambda_H}{\bar{\lambda}} = \frac{(1/3)4}{1.6} = 0.833. \quad (6)$$

Figure 2 illustrates how cumulative arrivals cluster even when the final counts are comparable.

F. Run Procedure

Before each run, the runner verifies that input and result queues are empty for three consecutive observations, exactly one worker is ready, the expected autoscaler is the only active policy, and its status is healthy. The system is idle for at least 30 seconds before the first arrival. The producer follows absolute monotonic deadlines and dispatches sends through a bounded thread pool, preventing one SQS network round trip from delaying later arrivals. It records scheduling lag when each sender starts its API call.

A collector long-polls the result queue while a monitor samples the Deployment, Pods, nodes, and both SQS queues

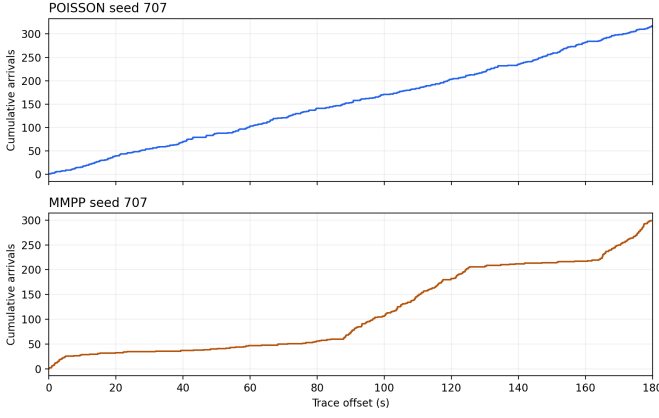


Figure 2. Cumulative arrivals for representative seed 707. Changes in MMPP slope expose alternating low- and high-rate periods.

Table III
MEASURED RUN TIMELINE

Step	Gate or observation
1	Remove previous policy and reset Deployment
2	Observe both queues empty three times
3	Apply one policy and await healthy status
4	Hold one ready, pre-pulled worker for 30 s
5	Replay 180-s trace on absolute deadlines
6	Collect results and sample state every 5 s
7	Mark last accepted input and last completion
8	Observe 15-s quiet window for duplicates
9	Observe scale-down and write final summary

every five seconds. Collection continues through a 15-second quiet window after all unique results arrive, which captures late duplicates. Configuration snapshots taken before and after the run detect changes to the Deployment or autoscaler specification.

Table III makes the measurement windows explicit. Producer deadlines use a local monotonic clock, but accepted arrival time comes from the SQS service response. Worker receipt and completion use the EKS node clock. Cross-clock subtraction is checked for impossible negative values. The monitor records desired, ready, available, and pending replicas; node readiness; restarts; and approximate queue counts independently of message processing.

A run is invalid if producer lag p95 exceeds 250 ms, monitoring has a gap longer than 15 seconds, a node becomes NotReady, a worker restarts, a Pod is persistently unschedulable, configuration changes, clock sanity checks fail, or any expected unique result is missing. Immediate post-run SQS counters are diagnostic because they are approximate. Failure to obtain a clean three-observation reset before the next run, a late result, or an unexpected experiment identifier is invalid. One automatic retry uses the same trace after both queues are purged and the SQS purge interval has elapsed.

G. Statistical Analysis

For metric Y , each seed and workload yields the paired ratio

$$R_i = \frac{Y_{i,KEDA}}{Y_{i,HPA}}. \quad (7)$$

The analysis reports all paired points, the median ratio, and the geometric mean ratio. A nonparametric paired bootstrap resamples complete ratios with replacement and uses the 2.5th and 97.5th percentiles for a 95% interval [13]. It also separates runs by treatment order as a sensitivity analysis. This avoids pseudoreplication from hundreds of correlated messages within one controller run.

The bootstrap uses 10,000 resamples and a fixed pseudo-random seed of 5120. The interval is the percentile interval, not bias-corrected and accelerated. Per-run percentiles use NumPy’s default linear interpolation. With 260–327 messages per run, p95 depends on roughly 13–16 upper-tail observations, another reason to avoid treating message-level values as independent evidence. The two primary endpoints were prespecified; no multiplicity-adjusted claims are made for secondary outcomes.

For the common seeds completed in both workload strata, a descriptive interaction is

$$I_i = \frac{R_{i,MMPP}}{R_{i,Poisson}}. \quad (8)$$

Values above one indicate that the KEDA/HPA ratio increased under MMPP. Because order and campaign time are not independently balanced in the realized extension, this quantity supports mechanism discussion rather than a causal workload-interaction claim.

Queueing relations provide consistency checks rather than a stationary model of the experiment. For c replicas with service rate μ , the reference utilization is $\rho = \lambda/(c\mu)$. Little’s Law, $L = \lambda W$, is interpreted cautiously because MMPP arrivals, changing capacity, and finite traces make the system transient.

IV. IMPLEMENTATION AND REPRODUCIBILITY

The worker is a non-root Python container. It long-polls one SQS message, validates the protocol, consumes the requested CPU time, sends a structured result, and then deletes the input. The result contains SQS acceptance time, worker receipt and completion times, Pod and node names, receive count, service demand, and a work digest. CPU time is measured with `time.process_time()`, so cgroup throttling increases wall time without reducing requested CPU work.

Each Pod processes at most one message at a time. It uses 10-second SQS long polling, requests one message, and sets a 120-second visibility timeout. The result is sent before input deletion, preserving SQS’s at-least-once semantics; the experiment identifier and job identifier allow the collector to distinguish duplicates from expected results. The boto3 client uses standard retry mode with up to eight attempts. These details affect queue residence and are held constant across policies.

AWS resources are defined by CloudFormation and `eksctl`. CloudFormation creates two SQS queues, one immutable ECR repository, and separate managed policies for

the worker and KEDA. The cluster uses one managed node group, public networking, and no NAT Gateway. The image tag includes a hash of the complete worker build context, and the resolved image URI is retained in each run summary.

The measured environment used EKS Kubernetes 1.34.10 and KEDA 2.17.2. Metrics Server was installed as the managed EKS add-on. Worker resources, health probes, autoscaler manifests, queue URLs, image tag, and resolved worker image are captured by the deployment pipeline or run summary. The image was pre-pulled before measured execution so treatment order did not determine whether the node had to download it.

Creation scripts require `ALLOW_AWS_CHARGES=YES`. Cleanup independently requires `ALLOW_AWS_DELETE=YES`, verifies the target account and region, deletes IRSA roles before the cluster and foundation stack, and audits EKS, EC2, EBS, NAT, Elastic IP, load balancer, CloudFormation, SQS, ECR, IAM, and log resources. The design keeps the expected session cost below USD 2 and the explicit ceiling at USD 20.

The repository includes unit tests for deterministic traces, workload bounds, message validation, temporal integration, manifest parsing, paired analysis, and the sequential stopping decision. The worker image, CloudFormation template, and `eksctl` configuration are also validated before cloud creation.

A. Retained Artifacts and Auditability

Each completed attempt directory contains producer events, accepted result messages, Kubernetes and SQS monitor samples, duplicate records, and a machine-readable summary. Immutable traces are retained separately by pattern and seed. Analysis discovers summaries recursively but accepts only records marked as main-campaign and valid. It then joins policies by workload and seed, so an interrupted or unpaired attempt cannot silently enter a paired estimate.

The analysis writes both tabular CSV outputs and publication figures. Its campaign-status record distinguishes planned runs, valid completed runs, complete pairs, and cells not executed. Reproduction therefore does not depend on console logs or values transcribed from this paper. The final lifecycle audit uses service-specific list operations instead of assuming that successful stack deletion implies complete cleanup.

V. RESULTS

A. Campaign Integrity

The measured campaign produced 35 valid main runs: 18 Poisson runs and 17 MMPP runs. These runs processed between 271 and 327 Poisson messages and between 260 and 310 MMPP messages. All expected messages produced exactly one accepted result. No run had a missing result, worker restart, NotReady node, persistent unschedulable Pod, or configuration change. Median producer lag p95 was 1.93 ms for Poisson and 2.77 ms for MMPP, far below the 250-ms validity threshold. The largest monitoring gap was 8.98 s, also below its 15-s threshold.

The first five pairs failed the predefined interval-precision criterion, so the campaign entered its five-seed extension.

Table IV
REALIZED MAIN CAMPAIGN

Campaign item	Count
Planned treatment cells	40
Valid completed cells	35
Complete Poisson pairs	9
Complete MMPP pairs	8
Valid but unpaired cells	1
Missing or duplicate accepted results	0
Worker restarts or NotReady nodes	0
Remaining AWS project resources	0

Table V
RETAINED WORKLOAD TRACE CHARACTERISTICS

Pattern	Traces	Jobs range	Mean rate	Median IAT CV
Poisson	9	271–327	1.627	0.991
MMPP	9	260–310	1.609	1.938

Extended cloud runtime and repeated short-lived login interruptions led to termination after nine complete Poisson pairs and eight complete MMPP pairs. One additional HPA/MMPP run had no KEDA counterpart and was excluded from all paired estimates. No value was imputed. Consequently, the results retain more replication than the initial stage but do not claim completion of the planned ten-pair maximum. The AWS cleanup audit subsequently found no remaining project resource.

Table V characterizes the nine generated traces of each pattern. The mean empirical rates were close, but median inter-arrival coefficient of variation was approximately twice as large for MMPP. Most MMPP messages arrived in the high state even though that state occupies one third of stationary time, as expected because arrivals are sampled more frequently while its rate is high.

B. Primary Outcomes

Table VI reports KEDA/HPA ratios. A ratio above one indicates a larger outcome under KEDA. For Poisson traces, the geometric-mean p95 latency ratio was 2.03 (95% bootstrap interval 1.44–2.81). For MMPP traces, it was 2.66 (1.56–4.73). HPA therefore had lower aggregate p95 latency under both arrival patterns in this configuration. These paired bootstrap uncertainty intervals exclude one, but the small realized sample and MMPP order confounding preclude a broad confirmatory claim.

The corresponding ready pod-seconds ratios were 0.522 (0.486–0.560) for Poisson and 0.701 (0.581–0.823) for MMPP. KEDA thus had 47.8% and 29.9% less ready-capacity exposure, respectively. The Poisson pod-seconds interval met the predefined plus-or-minus 10% precision criterion. The other three intervals did not. Figures 3 and 4 expose every paired point rather than only the aggregate estimates.

Direction counts reinforce the trade-off without treating messages as replicates. KEDA used fewer ready pod-seconds in all 17 pairs. HPA had lower p95 latency in 16 of 17

Table VI
PAIRED PRIMARY OUTCOMES, KEDA/HPA RATIO

Pattern	Outcome	<i>n</i>	Geo. mean	95% CI
Poisson	Latency p95	9	2.033	[1.436, 2.810]
Poisson	Pod-seconds	9	0.522	[0.486, 0.560]
MMPP	Latency p95	8	2.659	[1.558, 4.728]
MMPP	Pod-seconds	8	0.701	[0.581, 0.823]

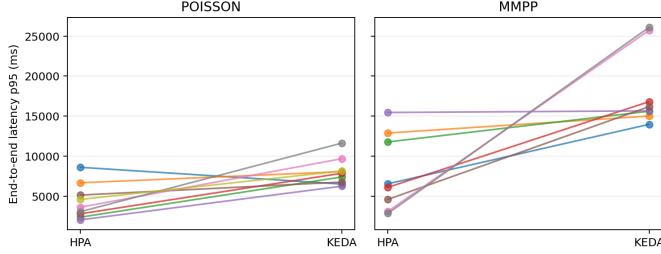


Figure 3. Per-trace p95 latency under HPA and KEDA. The dashed diagonal marks equal outcomes; points above it favor HPA.

pairs. The only exception was Poisson seed 101, for which the latency ratio was 0.757. Every MMPP ratio exceeded one, although seed 505 was nearly equal at 1.013. Thus, the aggregate directions were not produced by one isolated outlier, even though effect magnitude varied considerably.

C. Scaling and Backlog Behavior

Paired-only medians explain the primary trade-off without mixing the unpaired HPA/MMPP seed 909 into one policy. Under Poisson traffic, median p95 latency was 3.605 s for HPA and 7.845 s for KEDA, while median ready pod-seconds were 513.1 and 266.5. Median peak replica count was four for HPA and two for KEDA. Median maximum backlog was nine messages under HPA and 15 under KEDA. Median time from workload start until an additional Pod became ready was 36.8 s for HPA and 81.5 s for KEDA.

For the eight paired MMPP seeds, median p95 latency was 6.312 s for HPA and 15.907 s for KEDA, and median ready pod-seconds were 496.6 and 353.1. Both policies had a median peak of four replicas, but median maximum backlog was 16 for HPA and 39 for KEDA. Median workload-start-to-ready time was 46.8 and 54.3 s. This metric includes trace timing, sensing, reconciliation, scheduling, and readiness; it is not a direct measurement of controller polling delay.

Table VII reports exploratory mechanism ratios. Queue-wait p95 was 2.82 times larger with KEDA for Poisson and 3.01 times larger for MMPP. Integrated backlog ratios were 2.13 and 2.58. Queue residence therefore explains much of the end-to-end latency difference. KEDA reached lower peak capacity, especially under Poisson, while observed scale-out to a ready Pod occurred later on average.

The direction was again consistent. Integrated backlog favored HPA in all nine Poisson pairs and seven of eight MMPP pairs. Queue-wait p95 favored HPA in eight of nine Poisson and seven of eight MMPP pairs. KEDA reached

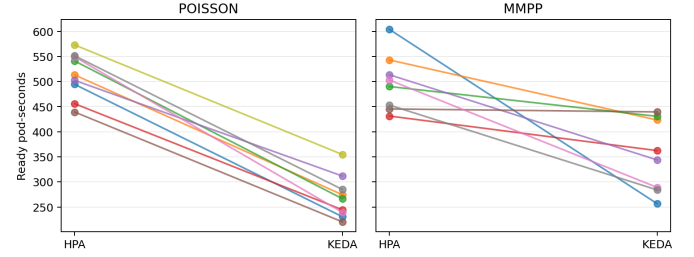


Figure 4. Per-trace ready pod-seconds. Points below the equal-outcome diagonal indicate lower ready-capacity exposure under KEDA.

Table VII
EXPLORATORY SECONDARY KEDA/HPA RATIOS

Outcome	Poisson	MMPP
Mean end-to-end latency	2.124	2.515
Queue-wait p95	2.816	3.006
Integrated backlog	2.134	2.580
Maximum sampled backlog	1.823	2.134
Workload-start-to-ready	1.675	1.370
Peak ready replicas	0.599	0.885

a lower peak replica count in all nine Poisson pairs; under MMPP it was lower in two and tied in six. These secondary comparisons were not multiplicity-adjusted and should be read as mechanism evidence rather than additional hypothesis tests.

Figure 5 shows that MMPP outcomes were heterogeneous. KEDA controlled maximum backlog on some traces but accumulated much larger peaks on others. This variation is consistent with sensitivity to the timing and length of high-state intervals, which equal message counts alone do not control.

D. Observed Control Timing and Replica Occupancy

Five-second monitor samples permit a coarse decomposition between a change in desired count and a subsequently observed ready Pod. Table VIII reports medians aligned to the first sample after workload start. Under Poisson, HPA first requested more than one replica after 31.6 s, while KEDA did so after 74.9 s. Desired-to-ready time was actually shorter for KEDA, indicating that the large total difference primarily preceded its scale request. Under MMPP, KEDA requested capacity 6.6 s later and had a longer observed desired-to-ready phase.

The difference column is computed within runs before taking the median, so it need not equal the difference between the preceding median columns. Sampling interval-censors all three values by roughly five seconds. Moreover, the data do not expose Metrics Server freshness, KEDA's exact poll instant, or internal HPA reconciliation separately. The safe conclusion is that KEDA's observed desired count changed later under Poisson, not that any single internal component caused the full delay.

Figure 6 directly links replica occupancy to pod-seconds. Poisson KEDA spent about 57.6% of the arrival window at one ready Pod and did not reach four in the mean occupancy

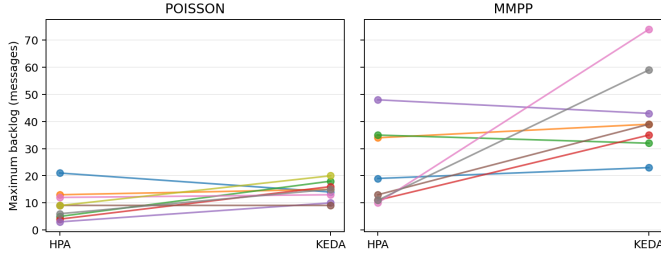


Figure 5. Per-trace maximum observed backlog. Lower points indicate stronger backlog containment during the finite workload.

Table VIII
MEDIAN SAMPLED SCALING TIMES FROM WORKLOAD START

Pattern	Policy	Desired > 1	Ready > 1	Difference
Poisson	HPA	31.6 s	35.1 s	10.0 s
Poisson	KEDA	74.9 s	80.0 s	5.2 s
MMPP	HPA	31.7 s	44.9 s	6.6 s
MMPP	KEDA	38.3 s	52.6 s	10.4 s

profile. Poisson HPA spent about 55.3% at four. MMPP drove both policies higher, but HPA still spent a larger share at four ready replicas. These observations explain why KEDA reduced ready-capacity exposure while allowing a larger queue.

Figure 7 presents one high-contrast pair rather than an aggregate. For MMPP seed 707, the same frozen trace produced an HPA p95 of 3.09 s and a KEDA p95 of 25.72 s. KEDA’s sampled backlog peaked at 74 messages versus 10 for HPA. The timeline shows repeated bursts, delayed desired-count changes, and the gap between desired and ready capacity. It is illustrative; the paired estimates continue to use all complete seeds.

E. Order and Workload Sensitivity

Treatment-order sensitivity was stable for ready pod-seconds but not for latency. For Poisson, geometric-mean pod-second ratios were 0.524 when HPA ran first and 0.519 when KEDA ran first. For MMPP they were 0.708 and 0.696. Geometric-mean latency ratios were 2.39 versus 1.66 for Poisson and 6.47 versus 1.56 for MMPP. Figure 8 makes the seed progression visible.

The MMPP schedule is critically limited: KEDA ran first for seeds 101–505, whereas HPA ran first for all complete extension seeds 606–808. The three late latency ratios were 3.53, 8.33, and 9.21. Treatment order, seed block, and campaign time are therefore completely confounded in that portion of the MMPP data. The overall MMPP latency ratio remains an accurate description of the executed pairs, but its magnitude cannot be assigned solely to autoscaler choice.

Across the eight seeds complete in both workload strata, the geometric-mean MMPP/Poisson ratio-of-ratios was 1.286 for p95 latency and 1.373 for ready pod-seconds. The respective uncertainty intervals were 0.709–2.211 and 1.158–1.618. Descriptively, burstiness reduced KEDA’s relative ready-capacity advantage and may have amplified its latency disadvantage,

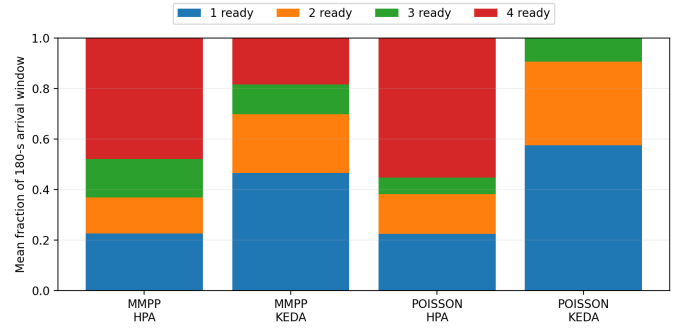


Figure 6. Mean fraction of the 180-second arrival window observed at each ready replica count. The distribution explains differences in ready pod-seconds.

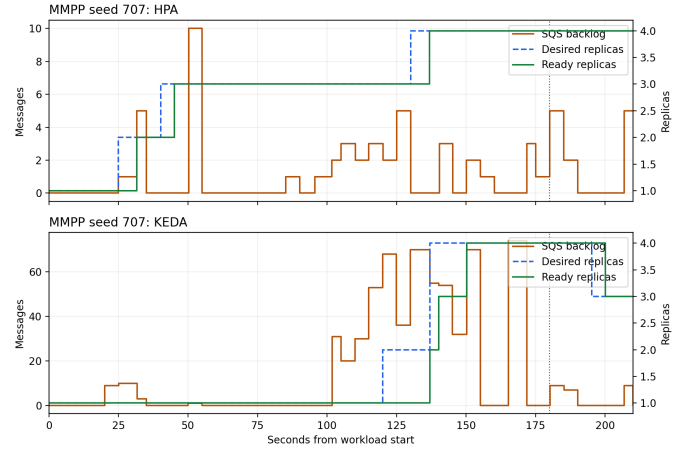


Figure 7. Backlog, desired replicas, and ready replicas for the MMPP seed 707 pair. The dotted vertical line marks the end of the 180-second arrival trace.

but latency heterogeneity and the order confounding prevent a causal interaction claim.

VI. DISCUSSION

A. Target Calibration Dominated Signal Proximity

The experiment does not support the intuitive claim that observing the queue necessarily improves tail latency. KEDA observed a signal closer to external demand, but the selected target of five messages per Pod preserved fewer active Pods during the mostly sub-capacity mean load. HPA’s 60% CPU target instead kept or requested more capacity once work reached a consumer. In this setting, that aggressive capacity response outweighed the informational advantage of the queue signal.

The nominal HPA calculation in Section III-C helps explain why. The 60% target is relative to the 200m request, not the 400m limit. Its 120m target utilization per replica is low compared with the 320m mean offered CPU demand, so three replicas are an idealized equilibrium before overhead. KEDA can remain at one replica until its sampled backlog justifies another. The observed Poisson desired-count times support this mechanism: KEDA’s larger delay occurred before the

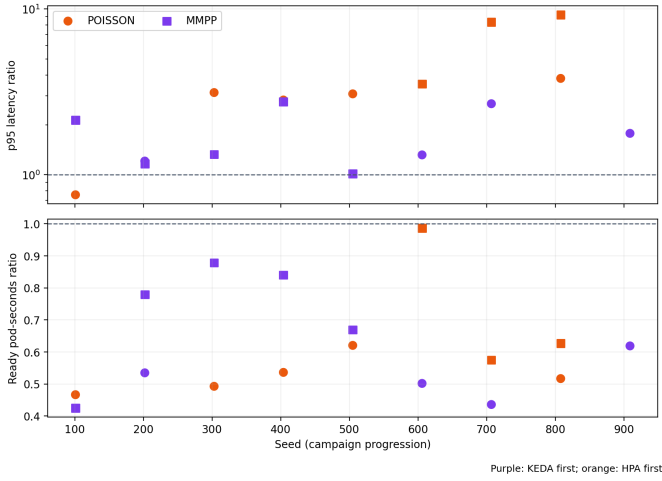


Figure 8. Primary KEDA/HPA ratios by seed and treatment order. Purple denotes KEDA first and orange HPA first; seed also tracks campaign progression.

Deployment requested a second Pod, not primarily while that Pod became ready.

During an MMPP high state, the arrival rate rises to 4 messages/s. With an idealized one-Pod service rate near 2 messages/s, at least two continuously busy Pods are needed merely to match new arrivals. Work accumulates while sensing and startup occur, and additional capacity must both serve new arrivals and drain the queue. This transient explanation is more appropriate than a stationary queueing model because capacity and arrival state both change during a 180-second trace.

B. A Ready-Capacity and Latency Trade-off

The trade-off was directionally consistent: all 17 ready pod-second contrasts favored KEDA, while 16 of 17 p95 latency contrasts favored HPA. This does not mean KEDA reduced the actual AWS bill. The EC2 node and EKS control plane were allocated for both treatments. Ready pod-seconds instead quantify how much consumer capacity the policy exposed within that fixed infrastructure. They can matter on clusters where freed requests permit other workloads or node consolidation, but neither consequence was measured here.

These findings apply to policy configurations, not controller brands in the abstract. A smaller KEDA backlog target, shorter polling interval, higher minimum replica count, or predictive activation could spend more pod-seconds to reduce waiting. Conversely, a higher HPA CPU target or longer stabilization could save capacity at a latency cost. The practical implication is that a queue-aware metric still requires a target aligned with a service-level objective; metric proximity alone does not guarantee responsiveness.

An operator choosing between these configurations should therefore start from an SLO and a capacity budget. The measured KEDA setup may suit throughput-first workloads that tolerate queueing and value lower ready-capacity exposure. The measured HPA setup better protected latency by maintaining more replicas. A stronger policy comparison

would tune HPA CPU targets and KEDA queue targets to equal latency or equal pod-second budgets, then compare the resulting Pareto front rather than assuming that equal replica bounds establish fairness.

C. Burstiness and Mechanism Uncertainty

Descriptively, MMPP reduced KEDA’s relative pod-second advantage from about 48% to 30%, while its latency ratio increased from 2.03 to 2.66. This pattern is plausible because high states pushed both policies toward the four-replica ceiling. It is not a clean causal workload effect: the ratio-of-ratios latency interval includes one, and late MMPP seeds are confounded with treatment order and campaign time.

The monitor establishes desired and ready replica timing, but cannot isolate every delay. Metrics Server freshness, KEDA polling, SQS counter approximation, HPA reconciliation, scheduler latency, and container readiness all remain combined or partially observed. Claims about those components are mechanisms consistent with the data, not direct component-level measurements. Future work should retain autoscaler events, HPA conditions, CPU samples, external metric values, scheduling timestamps, and image-start timestamps.

The results also illustrate why per-message significance tests would be misleading. Hundreds of messages in one trace share the same burst history, controller state, node, and Pods. Pairing at trace level yields only eight or nine independent contrasts per workload and appropriately exposes uncertainty and temporal sensitivity.

VII. THREATS TO VALIDITY

A. Internal Validity

The strongest internal threat is the realized MMPP order schedule. KEDA ran first for seeds 101–505 and HPA first for complete seeds 606–808. Order, seed block, and later campaign time cannot be separated. Sequential execution on one cluster also permits temporal variation in AWS control-plane latency, network conditions, node state, and background system Pods. Reset gates, pre-pulling, paired traces, and order reporting reduce but do not eliminate these effects.

The extension stopped with nine Poisson and eight MMPP pairs rather than ten of each. This non-protocol termination weakens precision and leaves one valid HPA run unpaired. The analysis excludes that run from every paired estimate and does not impute missing cells. Transparent exclusion prevents a fabricated pair but cannot recover the intended balance.

B. Construct Validity

Ready pod-seconds represent integrated ready replica count, not CPU work, energy, monetary cost, or useful throughput. A ready but idle Pod contributes equally to a busy Pod. Likewise, scale-out time begins at workload start, not at the unobserved instant when a true controller threshold is crossed. The sampled desired-count decomposition improves interpretation but remains interval censored.

SQS queue attributes are approximate and sampled every five seconds. Timestamp subtraction relies on AWS service

time and the EKS node clock; negative values beyond a small tolerance invalidate a run, but small positive clock offsets cannot be separated from queue wait. Pod-seconds use zero-order-hold integration between samples and are consequently discretized. Maximum backlog is the largest sampled approximate value, not the true continuous-time queue maximum.

Immediate post-run queue counters are retained but are not used alone as an invalidity condition because SQS exposes eventual approximations. Instead, the next run requires three consecutive empty observations after a bounded wait, and the collector rejects results carrying another experiment identifier.

C. Statistical Conclusion Validity

Eight or nine complete pairs provide limited bootstrap support, especially for wide MMPP effects. Percentile bootstrap intervals inherit the empirical sample's coarse tails and do not correct bias. Per-run p95 itself is estimated from only about 13–16 upper-tail messages. The sequential precision check and subsequent operational termination further complicate nominal interval interpretation. Individual points, direction counts, and sensitivity analyses make variation visible but do not create additional independent units.

Only latency p95 and ready pod-seconds were primary. Queue wait, backlog, replica timing, peak replicas, and interaction ratios are exploratory and correlated. Their intervals and direction counts receive no multiplicity correction and should not be interpreted as separate confirmatory tests.

D. External Validity and Reproducibility

The experiment uses one AWS region, one cluster, one node type, one CPU-bound worker implementation, standard SQS queues, one replica range, and one set of controller parameters. Results do not establish universal superiority. I/O-bound, batched, multithreaded, heterogeneous, or poison-message workloads may behave differently. The fixed node removes node-autoscaling variation but caps capacity; persistent unschedulability and node readiness failures were validity gates.

Conditioning MMPP traces on total count improves comparability of offered load but excludes unusually high- or low-count short realizations. The temporal clustering remains stochastic and recorded. Longer traces would include more state transitions and reduce dependence on a few bursts. KEDA was pinned, but managed EKS and add-on builds may evolve even with the retained Kubernetes version and manifests. Reproduction therefore requires reporting current cloud versions in addition to using the archived code and traces.

VIII. CONCLUSION

For the tested EKS worker and controller settings, CPU-based HPA provided lower end-to-end p95 latency, whereas SQS-based KEDA had fewer ready pod-seconds. KEDA/HPA geometric-mean latency ratios were 2.03 under Poisson and 2.66 under MMPP. Ready pod-seconds ratios were 0.522 and

0.701, equivalent to reductions of 47.8% and 29.9% in ready-capacity exposure, not direct monetary cost. KEDA also had larger median backlogs and later observed ready scale-out in this configuration, so queue visibility alone did not overcome the combined effects of target selection and control-path delay.

The result is a configuration-specific trade-off rather than a universal policy ranking. It demonstrates a reproducible way to compare internal resource pressure with external queued demand using frozen traces, run-level pairing, validity gates, retained raw observations, and audited cloud cleanup. Future work should vary the backlog and CPU targets, add longer traces and independent clusters, capture controller-internal metrics, and complete a balanced randomized campaign to separate treatment order from temporal drift. The strongest next comparison would estimate a latency-capacity Pareto front by tuning both policies to common SLO or pod-second budgets.

REFERENCES

- [1] Kubernetes Authors, "Horizontal pod autoscaling," accessed: 2026-08-31. [Online]. Available: <https://kubernetes.io/docs/concepts/workloads/autoscaling/horizontal-pod-autoscale/>
- [2] KEDA Authors, "AWS SQS Queue Scaler," accessed: 2026-08-31. [Online]. Available: <https://keda.sh/docs/2.17/scalers/aws-sqs/>
- [3] T.-T. Nguyen, Y.-J. Yeom, T. Kim, D.-H. Park, and S. Kim, "Horizontal pod autoscaling in Kubernetes for elastic container orchestration," *Sensors*, vol. 20, no. 16, p. 4621, 2020.
- [4] L. H. Phuc, L.-A. Phan, and T. Kim, "Traffic-aware horizontal pod autoscaler in Kubernetes-based edge computing infrastructure," *IEEE Access*, vol. 10, pp. 18 966–18 977, 2022.
- [5] L. Toka, G. Dobreff, B. Fodor, and B. Sonkoly, "Adaptive AI-based auto-scaling for Kubernetes," in *2020 20th IEEE/ACM International Symposium on Cluster, Cloud and Internet Computing (CCGRID)*, 2020, pp. 599–608.
- [6] D. Balla, C. Simon, and M. Maliosz, "Adaptive scaling of Kubernetes pods," in *NOMS 2020 – 2020 IEEE/IFIP Network Operations and Management Symposium*, 2020, pp. 1–5.
- [7] E. Casalicchio and V. Perciballi, "Auto-scaling of containers: The impact of relative and absolute metrics," in *2017 IEEE 2nd International Workshops on Foundations and Applications of Self* Systems (FAS*W)*, 2017, pp. 207–214.
- [8] M. Ezzeddine, F. Baude, and F. Huet, "Tail-latency aware and resource-efficient bin pack autoscaling for distributed event queues," in *Proceedings of the 14th International Conference on Cloud Computing and Services Science (CLOSER)*, 2024, pp. 50–64.
- [9] Amazon Web Services, "Available CloudWatch metrics for Amazon SQS," accessed: 2026-08-31. [Online]. Available: <https://docs.aws.amazon.com/AWSSimpleQueueService/latest/SQSDeveloperGuide/sqs-available-cloudwatch-metrics.html>
- [10] M.-Q. Tran, M.-V. Vu, and Y. Kim, "A survey of autoscaling in Kubernetes," in *2022 Thirteenth International Conference on Ubiquitous and Future Networks (ICUFN)*, 2022, pp. 263–265.
- [11] W. Fischer and K. Meier-Hellstern, "The markov-modulated poisson process (MMPP) cookbook," *Performance Evaluation*, vol. 18, no. 2, pp. 149–171, 1993.
- [12] Amazon Web Services, "IAM roles for service accounts," accessed: 2026-08-31. [Online]. Available: <https://docs.aws.amazon.com/eks/latest/userguide/iam-roles-for-service-accounts.html>
- [13] B. Efron and R. J. Tibshirani, *An Introduction to the Bootstrap*. Chapman and Hall/CRC, 1993.